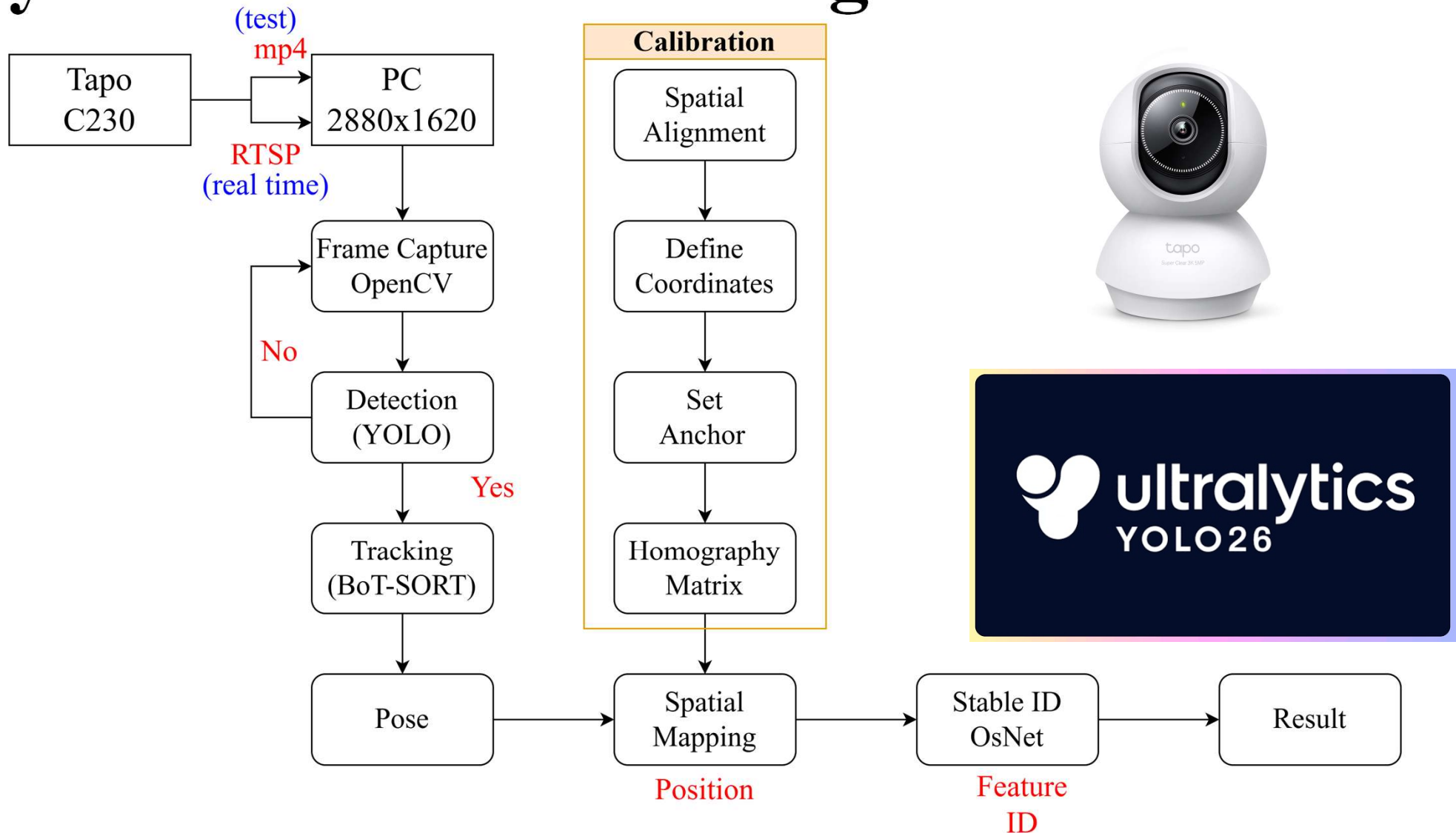


Person Detection & Tracking Results Summary

日期:2026/10/2 報告人：陳眾科

System Architecture Diagram



Detection(YOLO 26s 、 m 、 L)

只辨識COCO資料裡的Class 0 (人)



Ultralytics YOLO27

YOLO27 is coming soon

YOLO27 models are undergoing final R&D, with a launch anticipated later this year. **The models are not yet available, and no launch date has been set.** This page previews the upcoming models; features and benchmarks may change before release. Code examples below are intended for use once the models and package support are released and will not work with the current public package. Model weights, configurations, and implementation code are not being released at this time. Join the [YOLO27 waitlist](#) to be the first to hear when it launches.

YOLO26m	640	53.1	52.5	220.0 ± 1.4	4.7 ± 0.1	20.4
YOLO26l	640	55.0	54.4	286.2 ± 2.0	6.2 ± 0.2	24.8
YOLO26x	640	57.5	56.9	525.8 ± 4.0	11.8 ± 0.2	55.7

YOLO26-pose

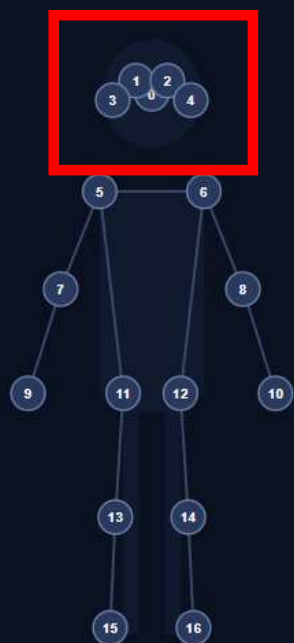
YOLO26-pose is pretrained on the **COCO Keypoints dataset**, which defines **17 keypoints** for human pose estimation. These keypoints cover the major joints and facial landmarks:

COCO-17 Keypoint Mapping

DATASET
COCO Keypoints

TOTAL POINTS
17

CATEGORIES
Face, Arms, Torso, Legs



0 Nose

2 Right Eye

4 Right Ear

6 Right Shoulder

8 Right Elbow

10 Right Wrist

12 Right Hip

14 Right Knee

16 Right Ankle

1 Left Eye

3 Left Ear

5 Left Shoulder

7 Left Elbow

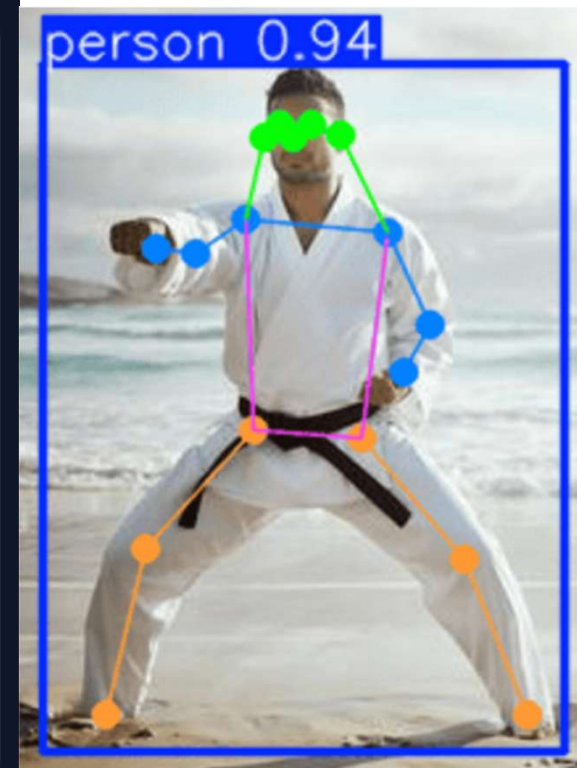
9 Left Wrist

11 Left Hip

13 Left Knee

15 Left Ankle

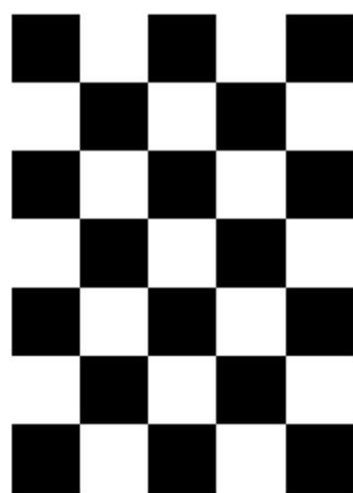
17 keypoints covering facial landmarks and major body joints — the standard mapping used by YOLO26-pose and COCO-format models.



<https://learnopencv.com/yolo26-pose-estimation-tutorial/>

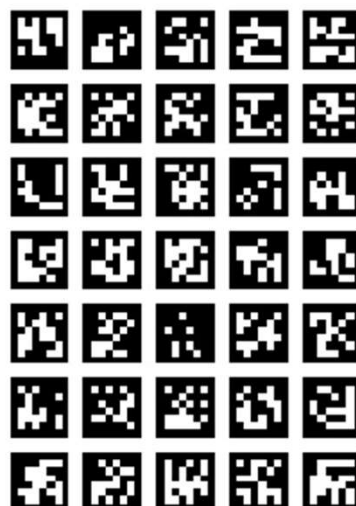
Calibration-(Spatial Alignment)

版本	檔案	作法	備註
v1	<code>calibration/homography_v1_manual.json</code>	手動點磁磚角 (<code>calibrate_boundary.py</code>)	點擊量測誤差約 8.9 cm
v2	<code>calibration/homography_v2_chessboard.json</code>	地板大棋盤自動角點	目前預設；點擊量測約 3.8 cm
v3	<code>calibration/homography_charuco.json</code>	地板 ChArUco 交點 (<code>calibrate_charuco_floor.py</code>)	試驗用，不取代 v2；板子附近約 2 cm。完整入鏡時與棋盤同屬平面 Homography



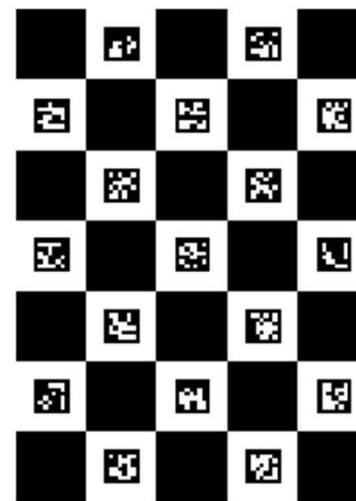
Chessboard

+



ArUco

=



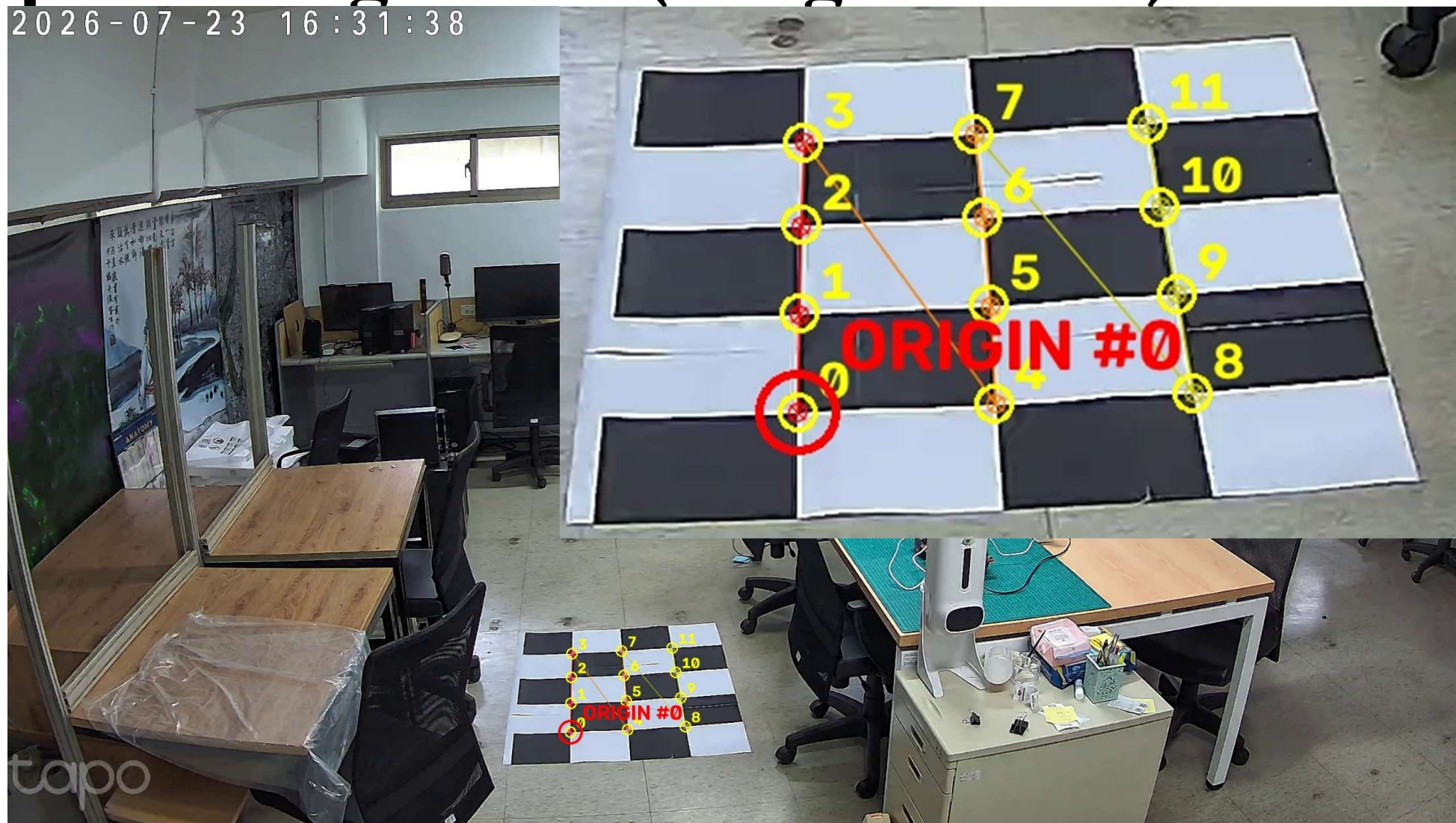
ChArUco

Spatial Alignment-(Manual)



Spatial Alignment-(Target-based)

2026-07-23 16:31:38



Spatial Alignment-(ChArUco)

2026-10-01 14:37:30



Different Angle

OLD chessboard 2026-07-23



NEW ChArUco 2026-09-30

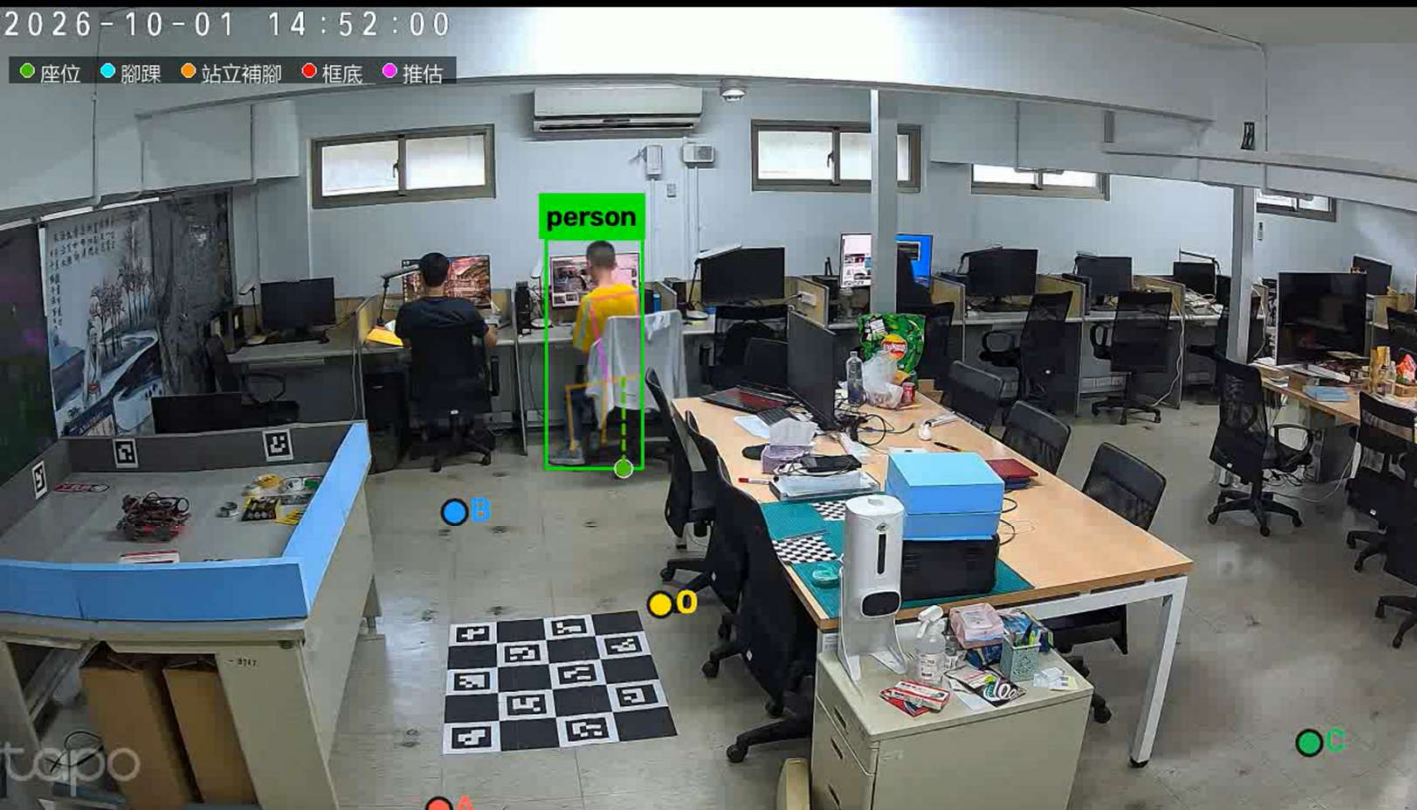


Result-(test6.mp4)

ChArUco + 5-pt comp test6

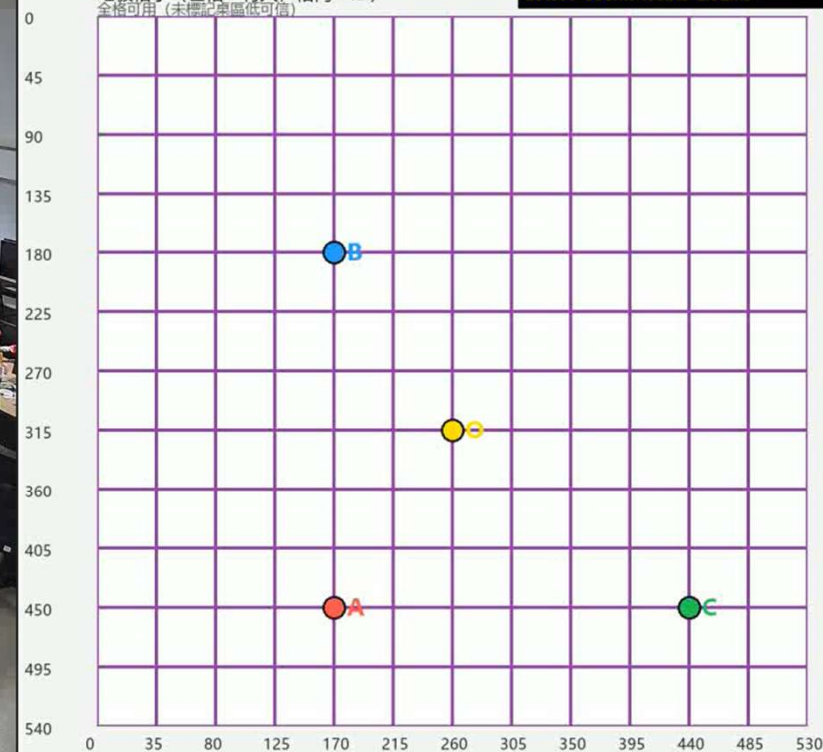
2026-10-01 14:52:00

● 座位 ● 腳踝 ● 站立補腳 ● 框底 ● 推估



地板格子 (色格=有人, 格內=ID)
全格可用 (未標記區域低可信)

detect 380ms locate 1.01ms

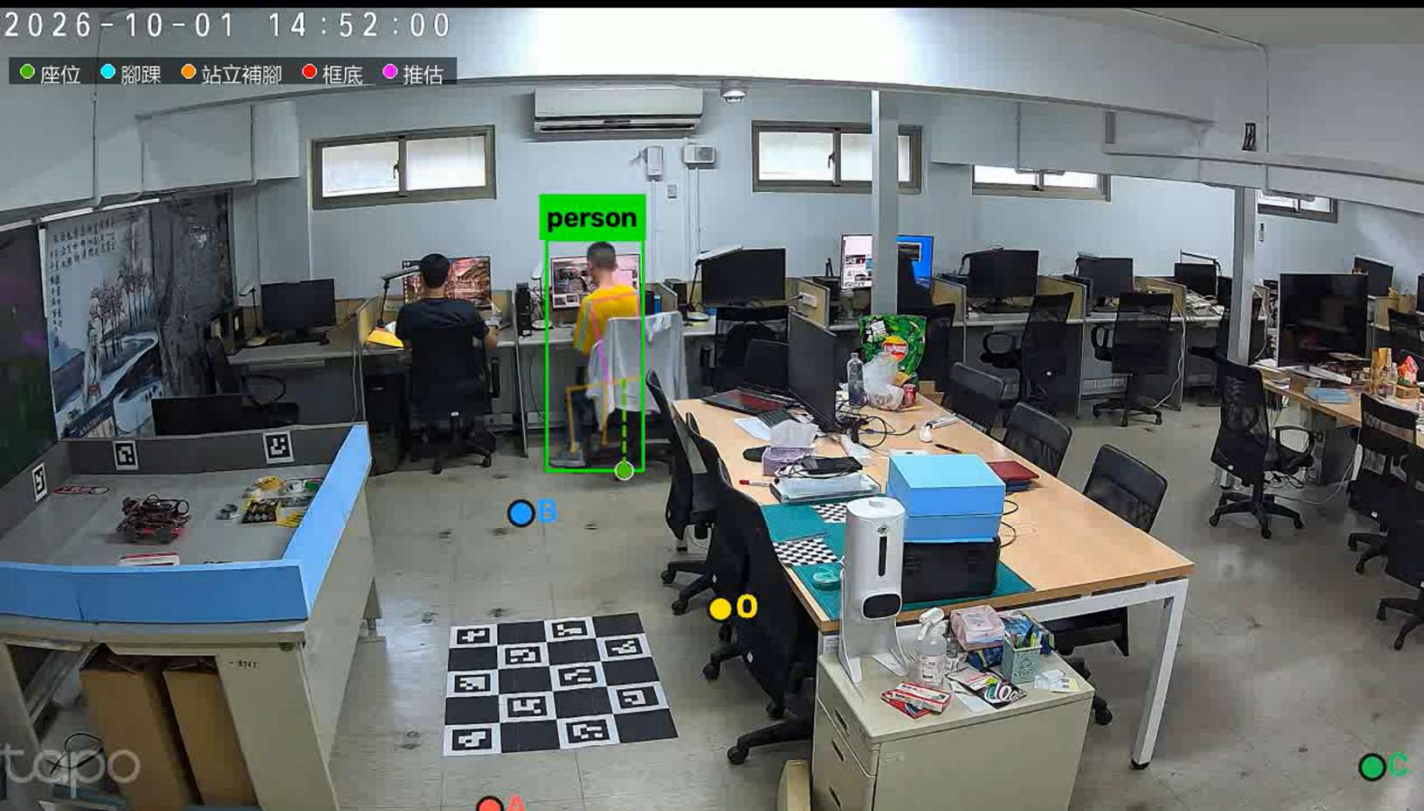


Result-(test6.mp4)

chessboard v2 + 4-pt comp test6

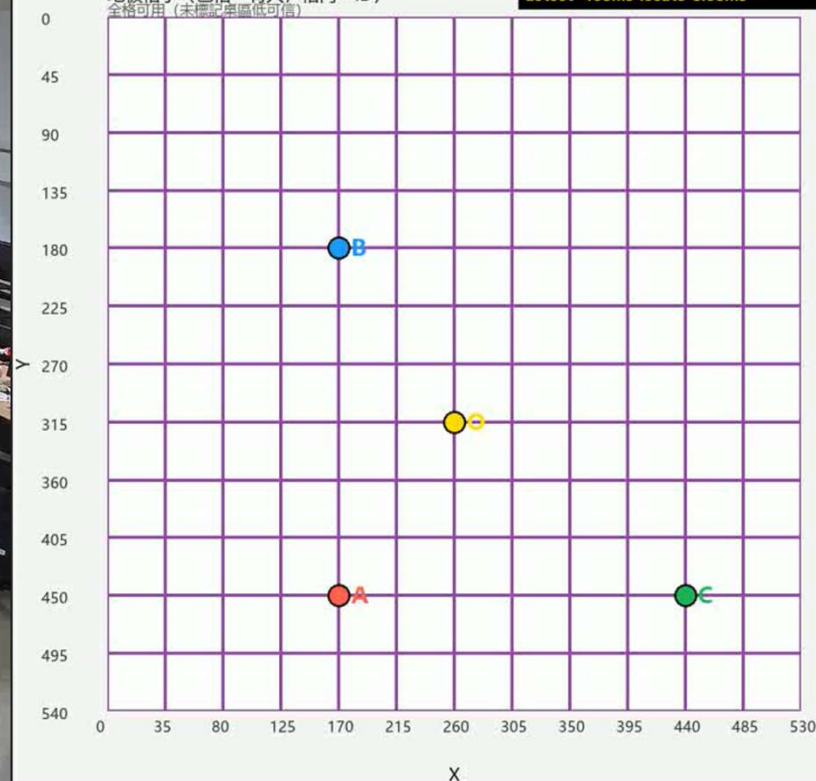
2026-10-01 14:52:00

● 座位 ● 腳蹠 ● 站立補腳 ● 框底 ● 推估



地板格子 (色格 = 有人, 格內 = ID)
全格可用 (未標記坐區低可信)

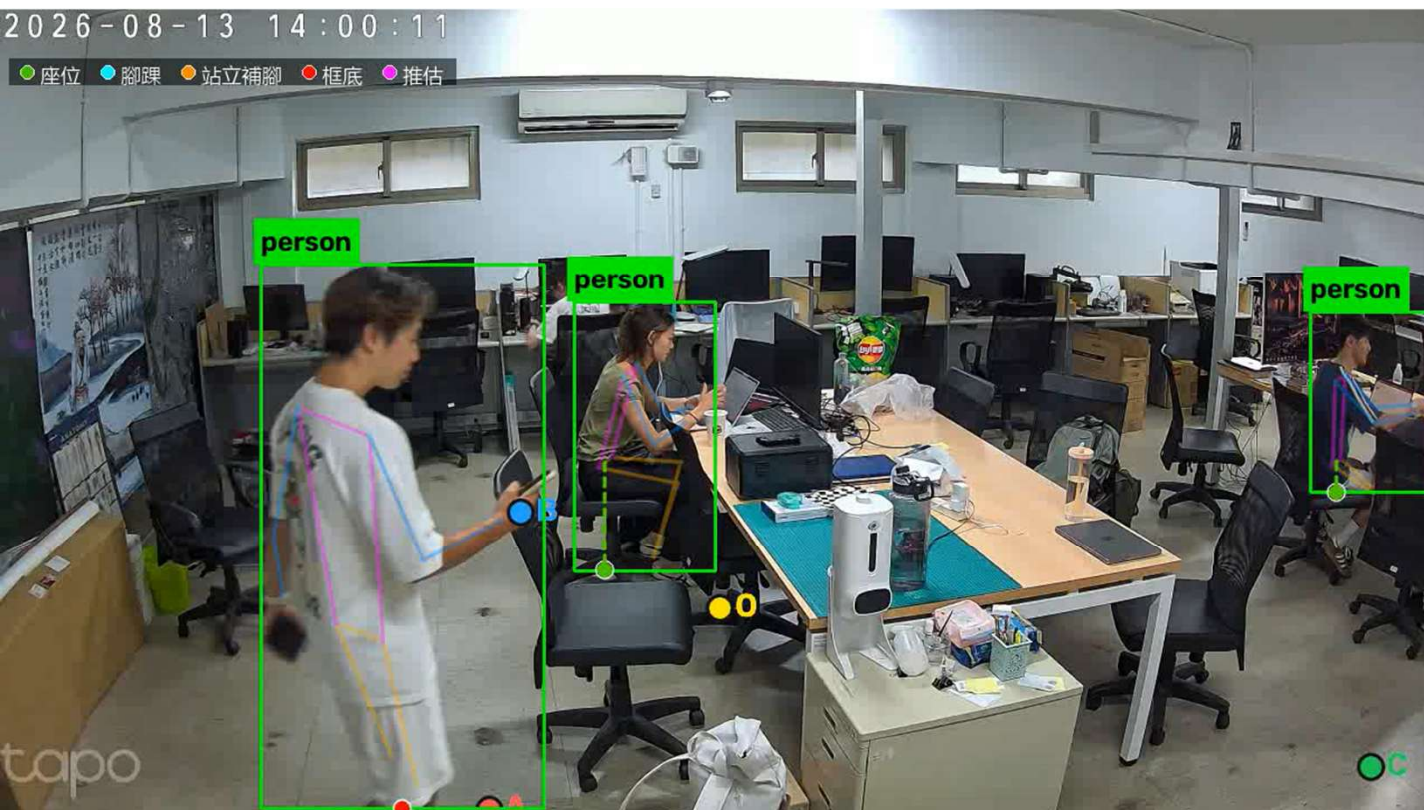
detect 463ms locate 3.65ms



Result-(test4.mp4)

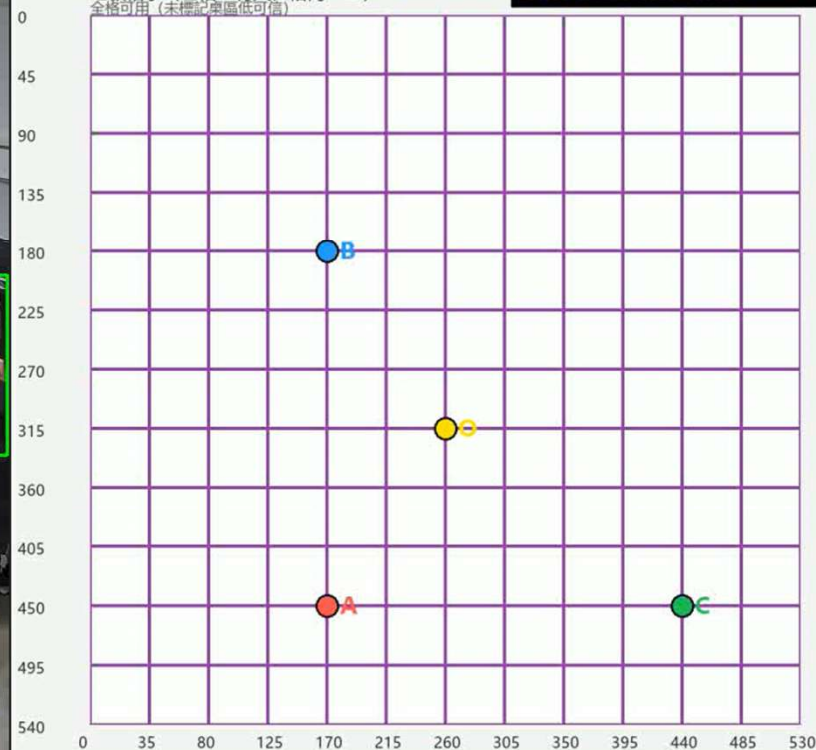
2026-08-13 14:00:11

● 座位 ● 腳踝 ● 站立補腳 ● 框底 ● 推估



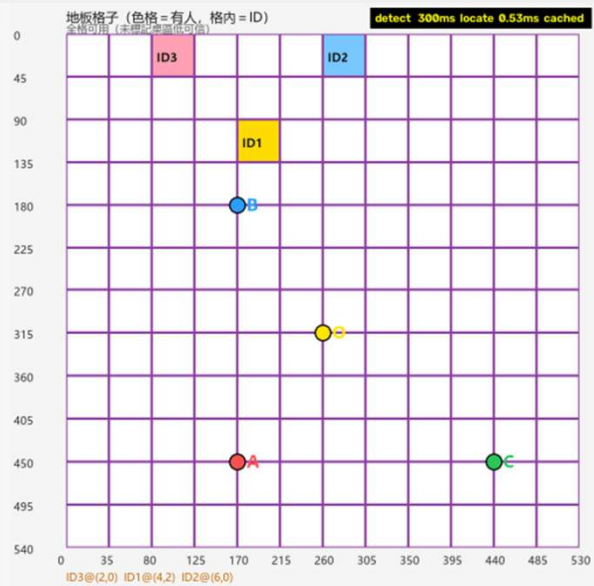
地板格子 (色格 = 有人, 格内 = ID)

detect 312ms locate 0.85ms



Result-(RTSP)

Grid 格子



Video 監視器



GUI

來源

☒ 本機影片 ☐ RTSP

影片 C:\5Gjump\test\test4.mp4

瀏覽...

校正 ☒ 棋盤 ☐ ChArUco 腳點 pose ☒ 易架 ☒ 追蹤 / ID ☒ 四點補償 ☒ A/B/C/O ☒ 少印 log ☐ 審查裁圖

YOLO yolo26s Re-ID osnet_ain

參數

conf 0.35
stride 5
cell-hold 2
min-hits 16
out-margin (cm) 45.0

將執行

```
python detect_grid.py --source C:\5Gjump\test\test4.mp4 --model yolo26s.pt --calib C:\5Gjump\calibration\homography.json --ref pose --conf 0.35 --stride 5 --cell-hold 2 --min-hits 16 --out-margin 45 --quiet --reid-model osnet_ain
```

Run

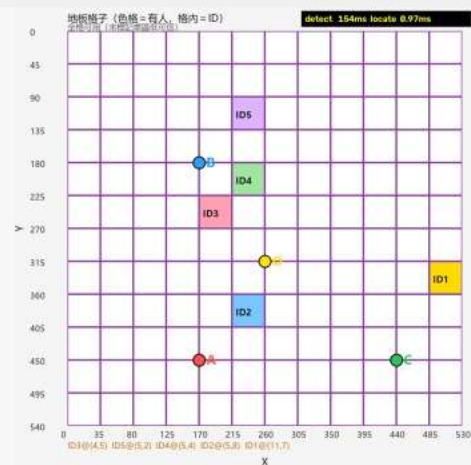
停止

暫停

本機影片可暫停，按停止結束

[ID-CHANGE] 11:29:47.266 elapsed= 62.88s video= 40.05s frame= 801 ID1,ID2,ID3,ID4 → ID1,ID2,ID3,ID4,ID5

Grid 格子



Video 監視器



感謝這三個月的指導